ZEPTO GROCERY SALES ANALYSIS

Project Workflow, Business Insights and Learning Report

1. Project Overview

This project involved analysing a grocery sales dataset through an end-to-end data analytics workflow. The objective was to prepare the data, answer business questions using SQL, and present the results through an interactive Power BI dashboard. The project provided practical experience in data cleaning, database handling, SQL-based analysis, and business intelligence.

2. Dataset Overview

The workbook contained 8,523 records and 12 columns. The fields included product information, item fat content, outlet details, outlet location and type, item visibility, item weight, sales, and ratings.

The dataset also contained missing values in the Item Weight column (1,452 records). No duplicate rows were found in the raw worksheet. These missing values were identified during inspection; this report does not assume they were imputed or removed.

3. Project Workflow

Step 1 – Initial cleaning in Excel using Copilot

Excel Copilot was used for initial data-cleaning tasks, including standardising inconsistent category values. For example, the Item Fat Content field contained variations such as “reg” and “Regular”, as well as “low fat” and “Low Fat”. These variations needed to be consolidated so that the categories could be analysed consistently.

Step 2 – Further cleaning using Python and Pandas

Pandas was used to handle the category standardisation that was not resolved through Copilot. A mapping was applied to convert “reg” to “Regular” and “low fat” to “Low Fat”. The column headers were also converted to lowercase and spaces were replaced with underscores, making the field names easier to use in SQL and Python.

The cleaned worksheet contained the same 8,523 records and 12 columns, with the fat-content categories standardised to “Low Fat” and “Regular”.

Step 3 – Loading data into PostgreSQL

The cleaned data was loaded into a PostgreSQL database named “zepto_grocery_database” and stored in a table named “zepto_grocery_table”. This provided a structured database source for querying the dataset.

Step 4 – SQL business analysis

SQL queries were used to answer business questions about outlet sales, product performance, location tiers, fat-content categories, and outlet ratings. The questions focused on extracting useful comparisons from the data.

Step 5 – Power BI dashboard

The PostgreSQL table was connected to Power BI and used as the source for the dashboard. The dashboard brought the analysis into a visual format to support comparisons and make the sales information easier to interpret.

4. Business Questions Addressed

1. What is the total sales generated by each outlet type?

	outlet_type text	total_sales double precision
1	Supermarket Typ...	787549.8928000013
2	Grocery Store	151939.149
3	Supermarket Typ...	131477.77639999994
4	Supermarket Typ...	130714.67460000006

2. Which product item types generate the highest total sales?

	item_type text	total_sales double precision	average_rating numeric
1	Fruits and Vegetabl...	178124.08099999995	3.96
2	Snack Foods	175433.9224000002	3.95
3	Household	135976.52539999998	4.00
4	Frozen Foods	118558.88140000009	3.97
5	Dairy	101276.46159999995	3.97
6	Canned	90706.72899999999	3.99
7	Baking Goods	81894.73640000001	3.98
8	Health and Hygiene	68025.83880000001	3.99
9	Meat	59449.86379999999	4.02
10	Soft Drinks	58514.16699999999	3.92
11	Breads	35379.119800000015	3.88
12	Hard Drinks	29334.680599999996	3.91
13	Others	22451.8916	3.95
14	Starchy Foods	21880.02739999999	3.92
15	Breakfast	15596.696600000001	3.93
16	Seafood	9077.869999999999	3.96

3. How do sales perform across different outlet location tiers?

	outlet_location_type text	total_products_sold bigint	total_sales double precision	average_sales numeric
1	Tier 3	3350	472133.03319999954	140.94
2	Tier 2	2785	393150.64759999956	141.17
3	Tier 1	2388	336397.81199999945	140.87

4. Do Low Fat items sell better than Regular items?

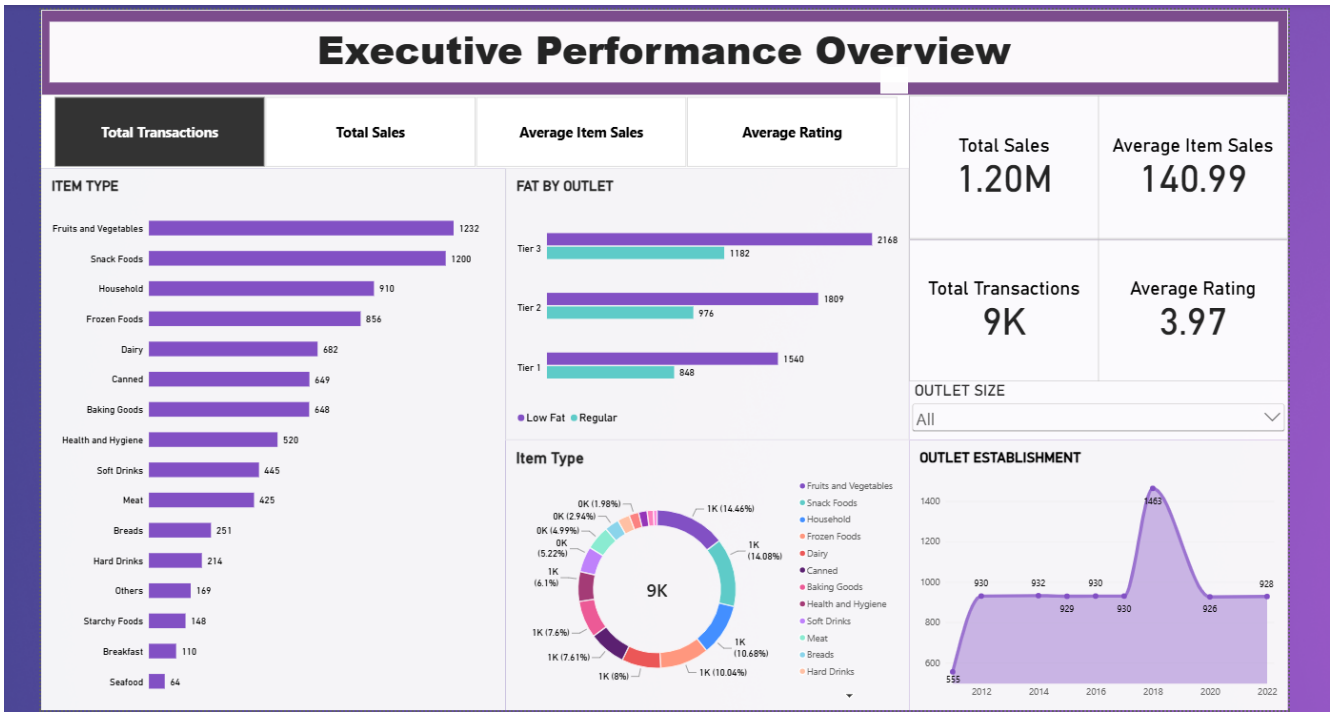
	item_fat_content text	item_count bigint	total_sales double precision	average_sales numeric
1	Low Fat	5517	776319.6884000006	140.71
2	Regular	3006	425361.8043999995	141.50

5. Which individual outlets have the highest customer ratings and sales performance?

1	OUT035	Supermarket Typ...	Tier 2	133103.9069999999	3.94
2	OUT046	Supermarket Typ...	Tier 1	132113.36980000007	3.96
3	OUT013	Supermarket Typ...	Tier 3	131809.01560000007	3.95
4	OUT018	Supermarket Typ...	Tier 3	131477.77639999994	3.97
5	OUT045	Supermarket Typ...	Tier 2	130942.7802	3.96
6	OUT027	Supermarket Typ...	Tier 3	130714.67460000006	3.95
7	OUT049	Supermarket Typ...	Tier 1	130476.85979999998	3.99
8	OUT017	Supermarket Typ...	Tier 2	129103.96039999987	3.98
9	OUT010	Grocery Store	Tier 3	78131.56659999998	3.98
10	OUT019	Grocery Store	Tier 1	73807.58239999998	4.00

5. Key Findings from the Dataset

- Total sales across the 8,523 records were approximately 1,201,681.49 (in the dataset's sales units).
- Supermarket Type1 generated the highest total sales among outlet types, at approximately 787,549.89.
- Fruits and Vegetables generated the highest total sales among item types, at approximately 178,124.08, followed closely by Snack Foods.
- Tier 3 outlets recorded the highest total sales by location tier, at approximately 472,133.03. Tier 1 recorded approximately 336,397.81.
- Low Fat items generated approximately 776,319.69 in sales, compared with approximately 425,361.80 for Regular items. This is total sales, not proof that an individual Low Fat item sells more.
- OUT035 recorded the highest total sales among the individual outlets, at approximately 133,103.91.
- Average ratings across outlet types and locations were relatively close, generally around 3.95 to 4.00.



6. Business Intelligence and Interpretation

The analysis suggests that Supermarket Type1 outlets are the largest sales contributors in this dataset. Fruits and Vegetables and Snack Foods are leading item categories by total sales. Tier 3 locations contribute the highest total sales, while Low Fat products account for a larger share of total sales than Regular products. These results can help a business compare outlet formats, product categories, and location tiers when reviewing sales performance.

Total sales should be interpreted alongside the number of records and average sales. A category with more records may generate higher total sales without having higher sales per record. Similarly, the dataset supports comparisons, but it does not by itself establish why a product or outlet performs better.

7. What I Learned

- How to identify inconsistent categorical values and standardise them before analysis.
- How to use Excel Copilot for initial cleaning and Pandas for more controlled, repeatable transformations.
- How to inspect a dataset, review missing values, check duplicates, and prepare consistent column names.
- How to load a cleaned DataFrame into PostgreSQL and use a database table as an analysis source.
- How to translate business questions into SQL queries using filtering, grouping, aggregation, sorting, and comparisons.
- How to connect PostgreSQL to Power BI and turn query-based analysis into a visual dashboard.
- How to interpret sales metrics carefully and distinguish total sales from average sales and other performance measures.

8. Conclusion

This project helped me practise a complete analytics workflow, from raw data preparation to dashboard creation. Using Excel Copilot, Pandas, PostgreSQL, SQL, and Power BI showed how different tools can work together in a single project. More importantly, it helped me understand how to move beyond cleaning data and writing queries towards answering business questions and communicating findings that can support business decisions.

Tools Used: Excel Copilot | Python (Pandas) | PostgreSQL | SQL | Power BI